

PCCST501: Computer Networks

Assignment 3 | Module 3: Data-Link Layer, Wireless LANs and Mobile IP

LAN, Ethernet and Wireless Network Investigation

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Course	PCCST501 Computer Networks (CO3)
Programme	B.Tech Computer Science and Engineering
Due date	17 August 2026

Q1. Data-link layer responsibilities; LLC vs MAC

The data-link layer sits above the physical layer and below the network layer. Its job is to move frames reliably between two nodes that share a link or a LAN, so the network layer can ignore the details of the wire or radio underneath.

- **Framing:** the bit stream is cut into frames with a header and trailer, so the receiver knows where each frame starts and ends.
- **Physical addressing:** MAC addresses name the sending and receiving interface on the local link.
- **Error detection:** a checksum (CRC) in the trailer lets the receiver reject damaged frames.
- **Flow control:** stops a fast sender from flooding a slow receiver.
- **Medium access control:** decides who may transmit when many stations share one channel.

The layer is split into two sub-layers (IEEE 802):

Aspect	LLC (IEEE 802.2)	MAC (IEEE 802.3 / 802.11)
Place in the stack	Upper half; talks to the network layer	Lower half; talks to the physical layer
Main role	Tells which network protocol is carried; optional acknowledgement and flow control	Builds frames, adds MAC addresses and FCS, controls access to the medium
Depends on medium?	No, same for all LAN types	Yes, different for Ethernet and Wi-Fi
Example	Marks the payload as IP or ARP	Ethernet uses CSMA/CD, Wi-Fi uses CSMA/CA

Q2. Ethernet frame format



Figure 1: Ethernet II frame with field sizes. The preamble and SFD are handled by the hardware, so Wireshark shows the frame from the destination MAC onwards.

Field	Size	Purpose
Preamble	7 B	Alternating 1s and 0s so the receiver can lock its clock to the sender.
SFD	1 B	Pattern 10101011 that signals the real frame begins.
Destination MAC	6 B	Address of the receiving NIC. It may be unicast, multicast or the broadcast address FF:FF:FF:FF:FF:FF.
Source MAC	6 B	Address of the sending NIC. Switches read it to learn where a host is.
Type / Length	2 B	A value of 0x0600 or higher is an EtherType naming the payload (0x0800 IPv4, 0x0806 ARP, 0x86DD IPv6); a smaller value is the payload length.
Payload	46 to 1500 B	Data from the upper layer. Short data is padded to 46 B.

Field	Size	Purpose
FCS	4 B	CRC-32 over the header and payload; if it does not match, the frame is dropped.

Frame size: 14 B header + 46 to 1500 B payload + 4 B FCS gives 64 to 1518 bytes (no VLAN tag).

Q3. Wireshark capture: MAC addresses from five frames

Steps to follow on your own computer:

- Open Wireshark and double-click the interface that shows live traffic (Wi-Fi or Ethernet).
- Run `ping 8.8.8.8`, open one website and run `arp -a` in a terminal to produce different frames.
- Stop the capture after about 30 seconds. Use the display filter `arp || icmp || dns`.
- Click a frame, expand *Ethernet II* in the details pane and note Destination, Source and Type.

Fill this table from your capture (rows are examples of what to look for, not real values):

Frame	Protocol	Source MAC	Destination MAC	EtherType
1	ARP request	(your PC)	ff:ff:ff:ff:ff:ff	0x0806
2	ARP reply	(router)	(your PC)	0x0806
3	ICMP echo request	(your PC)	(router / gateway)	0x0800
4	ICMP echo reply	(router)	(your PC)	0x0800
5	DNS query	(your PC)	(router / gateway)	0x0800

What to write as your observation: for traffic going out to the internet the destination MAC is the gateway's, not the far server's, because MAC addresses only work inside one link. ARP requests go to the broadcast address while replies are unicast. The first three bytes of a MAC (OUI) identify the manufacturer.

Q4. Switch operation and MAC address learning

A switch is a Layer-2 device that keeps a MAC address table (CAM table) mapping each MAC address to a port. It fills the table on its own by watching traffic:

- **Learn:** note the source MAC of every incoming frame against the arrival port (entries age out, usually after 300 s).
- **Forward:** if the destination MAC is in the table, send the frame out of that port only.
- **Flood:** if the destination is unknown or is a broadcast, send it out of all ports except the one it came in on.
- **Filter:** if the destination is on the arrival port itself, discard the frame.

Four hosts on one switch (MACs shortened for the diagram)

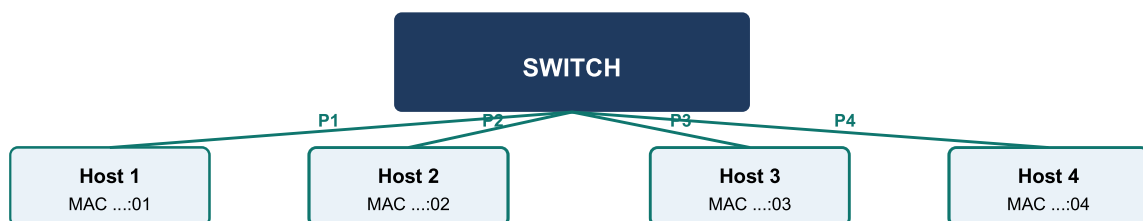


Figure 2: Example network used for the learning demonstration.

Step	Event	What the switch does	Table afterwards
0	Switch powered on	Nothing known yet	empty
1	Host 1 sends to Host 3	Learns Host 1 on P1; Host 3 unknown, so floods to P2, P3, P4	01:P1
2	Host 3 replies to Host 1	Learns Host 3 on P3; Host 1 known, so sends only to P1	01:P1, 03:P3
3	Host 2 sends to Host 3	Learns Host 2 on P2; sends only to P3	01:P1, 02:P2, 03:P3

Step	Event	What the switch does	Table afterwards
4	Host 4 sends to Host 1	Learns Host 4 on P4; sends only to P1	01:P1, 02:P2, 03:P3, 04:P4
5	Host 1 sends to Host 2	Both known; forwards to P2 only. Hosts 3 and 4 never see it	unchanged

On Cisco equipment the real table is shown with `show mac address-table`; in Packet Tracer you can follow the same steps in Simulation mode (add a screenshot if you try it).

Q5. Hub vs switch vs bridge

Parameter	Hub	Bridge	Switch
Layer	1 (Physical)	2 (Data link)	2 (some are Layer 3 capable)
Forwarding	Copies incoming bits to every other port	Forwards frames using a MAC table	Forwards frames using a MAC table, done in hardware
Collision domains	One for the whole hub	One per connected segment	One per port
Broadcast domains	One	One (without VLANs)	One per VLAN
Bandwidth	Shared by all devices	Shared inside each segment	Dedicated per port
Duplex	Half only	Half	Half or full
Practical use	Obsolete; maybe for traffic sniffing	Rare; joins two LAN segments	Standard in every modern LAN

Q6. CSMA/CD and why it matters less today

CSMA/CD (Carrier Sense Multiple Access with Collision Detection) controlled access on early shared-cable and hub-based Ethernet.

- **Listen first:** a station sends only when the line is quiet.
- **Listen while sending:** if the signal on the wire differs from what it is sending, a collision is happening.
- **Jam:** the station sends a short jam signal so everyone knows, then stops.
- **Back-off:** it waits a random number of slot times, chosen from 0 to $2^n - 1$ after the n -th collision (n capped at 10, give up after 16 tries), then retries.

The 64-byte minimum frame (slot time of 512 bit times) guarantees that the sender is still transmitting when news of a collision comes back from the far end of the cable.

Reasons its role has shrunk:

- Switches give each port its own collision domain, with one device per cable.
- Full-duplex links use separate wire pairs for sending and receiving, so a collision cannot happen and CSMA/CD is switched off.
- Gigabit and faster Ethernet is used almost entirely in full duplex.
- It still appears only with old hubs or when a duplex mismatch causes late collisions.

Q7. CSMA/CA in IEEE 802.11 and comparison with CSMA/CD

A radio cannot reliably listen while it transmits, and two stations can be out of each other's range yet both reach the access point (hidden terminal problem). Wi-Fi therefore avoids collisions instead of detecting them. The steps of the Distributed Coordination Function are:

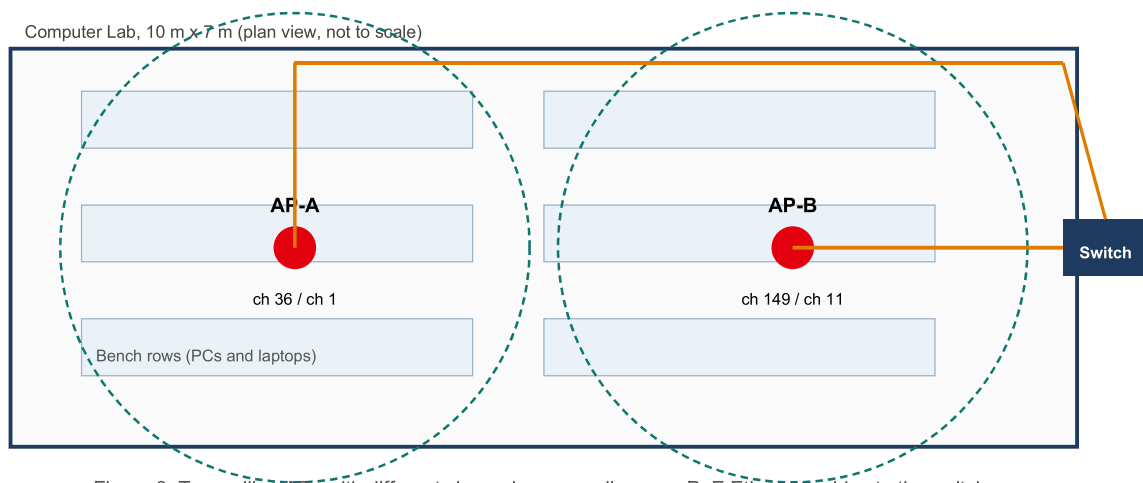
- Sense the channel (physically, and virtually using the NAV timer).
- If idle for a DIFS period, start; if busy, wait.
- Pick a random back-off counter that counts down only while the channel is idle.
- Send the frame; the receiver answers with an ACK after a SIFS gap. No ACK means a collision or loss, so the contention window is doubled and the frame is resent.

- Optionally exchange RTS and CTS frames; they set the NAV of all nearby stations and protect long frames from hidden nodes.

Point	CSMA/CD (Ethernet)	CSMA/CA (Wi-Fi)
Goal	Detect a collision and recover quickly	Prevent collisions before they occur
Medium	Wired	Wireless
ACK at MAC layer	No	Yes, for unicast frames
Back-off used	Only after a collision	Before each new transmission and after failures
Hidden terminal	Not an issue	Reduced with RTS/CTS and NAV
Overhead	Low	Higher (waiting times, ACKs)

Q8. Wireless LAN design for a college laboratory

Scenario: a computer lab of 10 m x 7 m with 30 desktop PCs on the wired network and about 45 wireless devices (student laptops and phones) at peak. Two access points cover the room.



Item	Design choice
AP placement	Two ceiling-mounted APs, 2.7 m high, one over each half of the room, about 3 m from the side walls. Cells overlap slightly in the middle for smooth roaming. Away from metal racks. PoE-powered from the switch.
Standard and band	Wi-Fi 6 (802.11ax), dual band. Clients are steered to 5 GHz; 2.4 GHz is kept for older devices.
SSIDs	CSE-Lab for students and staff, CSE-Guest for visitors (internet only, client isolation). The same SSID on both APs so devices roam.
Channels	2.4 GHz: AP-A channel 1, AP-B channel 11 (non-overlapping). 5 GHz: AP-A channel 36, AP-B channel 149, 20 or 40 MHz width to keep interference low.
Addressing	VLAN 10 (lab): 10.10.14.0/24, gateway 10.10.14.1, DHCP pool .20 to .240. VLAN 20 (guest): 10.10.20.0/24. VLAN 99 (management): 10.10.99.0/28, static IPs for APs and switch.
Wired side	PoE Gigabit switch, uplink to the college router/firewall with NAT.
Security	WPA3-Personal (or WPA2-AES) with a strong passphrase, or WPA2/WPA3-Enterprise with RADIUS. Disable WEP and WPS, change default admin passwords, update firmware, separate guest VLAN with rate limit. MAC filtering is optional only; it is not real security.
Capacity	About 22 wireless clients per AP, comfortably within what a Wi-Fi 6 AP handles.

Q9. Mobile IP and the need for mobility support

An IP address serves as both identity and location. If a laptop moves to another network, its old address no longer fits the new network's prefix, so packets still go to the old network and TCP connections break. Mobile IP (RFC

5944 for IPv4) lets a device keep one permanent address while moving.

Component	Role
Mobile Node (MN)	The device that moves between networks
Home Agent (HA)	Router on the home network that tracks the MN and forwards its packets
Foreign Agent (FA)	Router on the visited network that provides a care-of address
Home address	Permanent address of the MN
Care-of address (CoA)	Temporary address that shows the MN's current location

Operation:

- **Agent discovery:** agents send advertisements so the MN can tell it has moved.
- **Registration:** the MN registers its CoA with the HA, through the FA if there is one.
- **Tunnelling:** packets sent to the home address are caught by the HA, wrapped in a new IP header (IP-in-IP) addressed to the CoA, and unwrapped at the FA or MN.
- **Return traffic:** the MN replies directly to the sender, or uses reverse tunnelling when ingress filtering would block it. The indirect forward path is called triangle routing.

Why it is needed: phones, laptops and vehicles keep changing networks (Wi-Fi to mobile data, campus to home). Mobility support keeps calls, downloads and remote sessions running. Mobile IPv6 reduces the triangle routing delay through route optimisation.

Q10. Wired Ethernet frame vs wireless LAN frame

To see 802.11 frames, Wireshark needs an adapter in monitor mode (often an external USB adapter on Windows). If that is not possible, compare with a documented example, as below.

Dest MAC 6 B	Src MAC 6 B	Type 2 B	Payload 46 to 1500 B	FCS 4 B
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Figure 4a: Wired Ethernet II frame (excluding preamble and SFD)

Frame Ctrl 2 B	Duration 2 B	Addr 1 6 B	Addr 2 6 B	Addr 3 6 B	Seq Ctrl 2 B	Addr 4* 6 B	Frame body 0 to 2312 B	FCS 4 B
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Figure 4b: IEEE 802.11 data frame (*Addr 4 only in wireless distribution systems)

Feature	Ethernet II	802.11 WLAN
MAC header size	14 bytes	24 bytes (30 with Addr 4, plus 2 for QoS)
Addresses	2 (destination, source)	Up to 4, meaning set by the To-DS and From-DS bits
Frame types	One kind of frame	Management (beacon, probe, authentication), control (RTS, CTS, ACK) and data
Acknowledgement	None at MAC layer	Every unicast frame is acknowledged
Duration / NAV	Absent	Duration field supports virtual carrier sensing
Sequence control	Absent	Detects duplicates and handles fragments
Maximum payload	1500 B	About 2304 to 2312 B
Encryption	Not at Layer 2 (MACsec optional)	Built in (CCMP in WPA2, GCMP in WPA3); Protected flag in Frame Control
FCS	CRC-32, 4 B	CRC-32, 4 B

Conclusion: the wireless frame has more fields because the radio medium is shared and unreliable, so it needs extra addressing, timing, acknowledgement and security information that a switched wired LAN can do without.

Deliverables checklist

Deliverable	Where
Ethernet frame diagram	Q2, Figure 1
Wireshark screenshots	Q3 and Q10 (add your own)
Wireless LAN design	Q8, Figure 3 and table
Comparison tables	Q5, Q7, Q10
README.md	In the GitHub repository (separate file)

References

- Forouzan, B. A., *Data Communications and Networking*, McGraw-Hill.
- Tanenbaum, A. S. and Wetherall, D., *Computer Networks*, Pearson.
- IEEE 802.3 (Ethernet) and IEEE 802.11 (Wireless LAN) standards.
- RFC 5944, IP Mobility Support for IPv4, Revised.
- Wireshark User's Guide, <https://www.wireshark.org/docs/>